

Introduction

Engine9 is a domain-specific programming language designed for engineering calculations. It provides a simplified syntax for defining variables, evaluating expressions, and computing results using built-in functions.

Lexical Grammar

Tokens in Engine9 include:

- **Keywords:** let, print
- **Identifiers:** variable names (e.g., mass, velocity)
- **Literals:** integers and floating-point numbers
- **Operators:** +, -, *, /, =
- **Delimiters:** parentheses (,)

Syntactic Grammar

Basic structure:

```
statement → declaration | print_statement
declaration → let identifier = expression
print_statement → print identifier
expression → term ((+ | -) term)*
term → factor ((* | /) factor)*
factor → identifier | literal | function_call
function_call → identifier (arguments)
arguments → expression (, expression)*
```

Semantic Structure

- Variables must be declared before use
- Expressions must use valid operators
- Function calls must match defined built-in functions
- Division by zero is not allowed

Data Types

- Integer
- Float

Types are determined at runtime.

Reserved Words

- let
- print

Operators

- Arithmetic: +, -, *, /
- Assignment: =

Built-in Functions

- force(m, a)
- kinetic_energy(m, v)
- ohm_current(v, r)
- work(f, d)
- power(w, t)
- pressure(f, a)
- density(m, v)
- momentum(m, v)

Statement Structure

Variable Declaration

```
let x = 10
```

Expression

```
let result = x * y
```

Print Statement

print result

Example Program

```
let mass = 10
let accel = 9.8
let forceValue = force(mass, accel)
print forceValue
```

Implementation Details

Engine9 is implemented in Python using:

- Tokenizer (lexer)
- Parser
- AST representation
- Symbol table (hash table)
- Interpreter